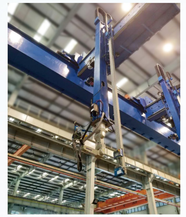
Application

SIASUN's medium and thick plate welding robots have garnered successful implementations across various welding domains. The SIASUN intelligent welding system, functioning as a flexible and autonomous welding processing unit, provides a solid foundation for mass-producing high-quality equipment in assembly line settings, ensuring streamlined manufacturing. It also enables the feasibility of highly adaptable short-term, flexible production, thereby enhancing responsiveness to dynamic market demands.



新松机器人自动化股份有限公司
SIASUN ROBOT & AUTOMATION CO., LTD.

Add:NO.33 Quanyun Road,Hunnan District,Shenyang,P.R.China.
Tel:400 800 8666
E-mail:market@siasun.com
WeChat：siasunrobot
Web：https://www.siasun.com/
VER：SX023004-V1.0

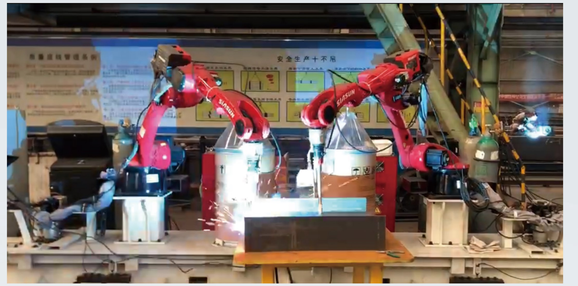
SIASUN
新松机器人

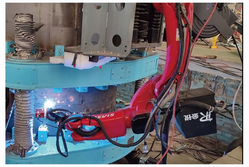
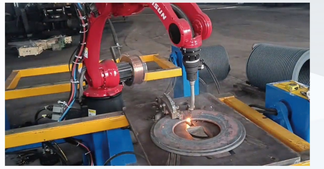
SIASUN MEDIUM AND THICK PLATE WELDING MACHINE SET

YOUR BEST WELDING ASSISTANT

M500

Application



SIASUN Welding Engineer System

Controller

Teach pendant

Welding torch
GL500N-500I,TRM602WHD

Welding machine
Dex2 500MPR

Wire feeder

Wire box

SIASUN SR-series robot equipped with a new intelligent welding control system,creating exclusive welding equipment for you to easily achieve welding automation

Welding machine parameters

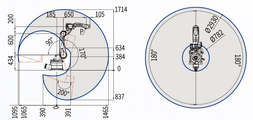
Model	Dex2 500MPR		
Rated input capacity	23.3KVA/21.4KW	Parameter channel storage	50
Rated output current	500A	Recirculating cooling tank	Anycool-66
Rated output voltage	39V	Rated power	370W
Rated duty cycle	DC 100%@450A	Rated voltage	380Vac
	Pulse100%350A	Working temperature	-10℃~40℃(Welding power supply can be started at -39.℃)
No-load voltage	77V	Appearance size	L*W*H (mm) 647*291*572
Efficiency	91%@500A	Weight	40kg
Output characteristic	CV	Shell protection level	IP23 S
Wire feeding speed	0.5-28m/min	Cooling method	Air cooled/Water ccooled.
Power factor	0.92	Cooling water flow rate	3.5L/min
Insulation level	H	Cooling water capacity	6.8L

Industrial robot

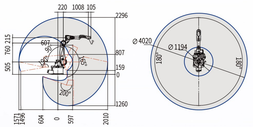
The SIASUN SR series industrial robots are specially designed for the welding field.
Streamlined appearance,high rigidity and lightweight body,combined with strong driving force,achieve ultimate speed and accuracy.

Type		SR12A-12/1.46	SR25A-12/2.01
Max payload		12kg	12kg
Repeatability		±0.03mm	±0.07mm
Max.reach		1465mm	2010mm
Motion Range	J1	±180°	±180°
	J2	+90°，-170°	+90°，-165°
	J3	+180°，-200°（combined）	+190°，-200°（combined）
		+150°，-90°（single）	+160°，-85°（single）
	J4	±180°	±180°
	J5	+145°，-160°	+145°，-160°
	J6	±360°	±360°
Maximum Speed	J1	200°/s	200°/s
	J2	200°/s	220°/s
	J3	220°/s	220°/s
	J4	400°/s	400°/s
	J5	430°/s	430°/s
	J6	720°/s	720°/s
Moment	J4	23N·m	23N·m
	J5	20N·m	17N·m
	J6	9N·m	9N·m
Inertia	J4	0.4kg·m²	0.4kg·m²
	J5	0.38kg·m²	0.35kg·m²
	J6	0.1kg·m²	0.1kg·m²
Weight		120kg	250kg
Protection（wrist）		IP65	IP65
Wiring method		built-in wiring	

SR12A-12/1.46



SR25A-12/2.01



Welding Fuction

Position Sensing Detection Function:

Contact Sensing Detection Function

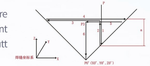
This primarily encompasses three-directional sensing, start point detection, end point detection, arc sensing, and contact probe sensing functionalities. It ensures that the welding process remains unaffected by deviations in weld seam positioning due to material preparation, assembly and fit-up welding, or clamping during welding setup. By automatically seeking out the weld seam location, it guarantees smooth and accurate welding. Contact sensing detection is achieved by the wire, nozzle, or probe coming into contact with the workpiece to sense its position. It is straightforward to operate and does not require additional sensing devices, thereby enhancing torch flexibility. Boasting high precision and excellent accessibility, this feature offers significant advantages.

Three-dimensional Sensing:

The three-directional sensing function typically senses the workpiece in three axes: x, y, and z. When the workpiece shifts, the robot can detect the deviation in the welding path and make the necessary adjustments accordingly.

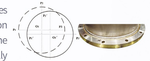
Start (End) Point Detection:

When the welding workpiece shifts within a certain range, the start (end) point detection feature can be utilized to re-anchor the correct starting (ending) point for the task. This start (end) point detection function is applicable to various types of weld seams, including fillet welds and butt joint grooves.



Arc Sensing:

The arc sensing function is tailored for workpieces with curved or arc-shaped profiles. It involves using the wire's contact with the workpiece to detect any deviation between the actual position of the workpiece and the taught position. Once this data is processed by the robot, it adjusts the original welding path accordingly. By comparing two sets of data, the robot automatically devises a new welding trajectory, thereby enabling the automated welding of misaligned workpieces.



Contact Probing Sensing:

The contact probing sensing function facilitates the detection of groove positions. By employing contact probing sensing, the general location of the weld seam is identified. Subsequently, in accordance with specific requirements, it integrates with other detection methods to precisely locate both the starting point and the weld seam position.

Non-Contact Sensing Detection Function

Non-contact Sensing Detection typically employs point or line laser sensors as tools, where the robot, through these sensors, detects the workpiece's positional data in the X, Y, and Z axes within the inspection coordinate system. This information, combined with an understanding of the workpiece's characteristics and weld seam type, is then used to determine the starting and ending points of the weld seam. In comparison to contact sensing, this method is capable of accommodating greater deviations in weld seam positioning and is not influenced by factors such as wire bending or inconsistent extension length.

Point Laser Sensing Detection:

By installing a point laser sensor around the welding torch and calibrating the point laser equipment, the relative positional relationship between the sensor and the welding torch's TCP (Tool Center Point) is established. The robot can then leverage sensor data to adjust and optimize the welding start and endpoints.

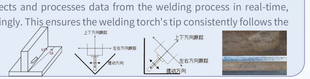
Line Laser Sensing Detection:

Equipping the welding torch with a line laser sensor, when the robot executes the welding program, it initially employs the line laser sensor to scan the starting area. The robot acquires the positional information of the weld seam's starting point via the line laser sensor, and based on the weld seam type, it automatically determines the precise location of the welding starting point.

Seam Tracking Function:

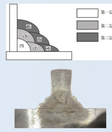
Arc Tracking Function

In instances where the actual weld seam trajectory deviates from the taught program path or the workpiece undergoes thermal deformation during welding, the robot collects and processes data from the welding process in real-time, adjusting and compensating the running path accordingly. This ensures the welding torch's tip consistently follows the weld seam.



Laser Tracking Function

In the event of weld seam displacement, the robot, utilizing a forward-mounted laser sensor, gathers and processes weld seam position data. This enables real-time correction of the motion path during the welding process and automatic endpoint detection. Line lasers can track and adjust for fillet welds (with or without bevel), butt welds (with or without bevel), lap welds, vertical welds, and overhead weld positions, catering to full-position welding tracking requirements.

Multi-Pass Welding Function:

When a single pass of welding fails to meet process requirements and multiple passes, even layers with multiple passes, are necessary, traditional robotic welding methods often fall short of productivity demands. To streamline and expedite the process, SIASUN has developed a multi-pass welding module.
This multi-layer and multi-pass welding capability can be integrated with other functions such as contact sensing detection and weld seam tracking. By leveraging contact sensing detection and weld seam tracking, information about the workpiece gathered during the initial layer's welding is recorded. After systematic processing and calculation, this information is directly applied to subsequent layers, ensuring welding quality. Meanwhile, the welding process parameters and torch orientation adjustments can be consistently applied across every layer. Employing multiple strategies to guarantee welding quality, this approach delivers satisfactory welding outcomes to customers.